

KOLPASHEVO, Russian Federation

WMO: 292310

Lat: **58.3231N**Lon: **82.9279E**Elev: **75**StdP: **100.43**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.7	(c) -36.3	(d) -41.9	(e) 0.1	(f) -39.3	(g) -38.9	(h) 0.1	(i) -36.1	(j) 5.9	(k) -12.8	(l) 5.3	(m) -12.1	(n) 0.8	(o) 300	(p) 0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.5	19.1	26.8	18.4	25.0	17.6	20.8	25.8	19.7	24.6	18.6	23.2	2.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.7	17.9	12.9	21.8	16.9	12.1	21.0	60.7	25.9	56.8	24.8	53.4	23.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.1	4.4	-42.6	31.5	3.7	1.6	-45.2	32.7	-47.4	33.6	-49.4	34.6	-52.1	35.7
			-41.6	23.1	3.4	1.8	-44.1	24.4	-46.1	25.5	-48.0	26.5	-50.5	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-20.4	-16.0	-7.1	0.9	8.7	16.0	18.2	14.9	8.6	1.0	-10.2	-16.9
	DBStd	14.67	9.27	8.04	6.63	6.02	5.80	4.63	3.45	3.66	4.46	5.21	8.66	9.25
	HDD10.0	4387	941	729	531	277	97	8	0	5	79	281	604	833
	HDD18.3	6816	1199	962	789	524	304	96	46	117	294	538	854	1091
	CDD10.0	694	0	0	0	3	56	188	253	156	36	2	0	0
	CDD18.3	81	0	0	0	0	4	26	41	10	1	0	0	0
	CDH23.3	763	0	0	0	0	55	262	354	86	6	0	0	0
	CDH26.7	152	0	0	0	0	11	57	72	12	0	0	0	0
Wind	WSAvg	2.0	1.8	2.0	2.2	2.3	2.4	1.9	1.6	1.7	1.9	2.2	2.2	2.1
Precipitation	PrecAvg	514	24	18	20	26	51	59	72	77	51	46	39	32
	PrecMax	733	66	38	50	55	129	115	203	179	106	97	77	74
	PrecMin	360	3	2	2	4	16	7	0	21	10	19	10	7
	PrecStd	88	11	8	10	12	25	29	46	36	20	17	13	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	1.1	8.2	18.5	28.1	30.5	30.6	28.5	24.1	16.0	5.2	0.7
		MCWB	-1.5	0.0	3.1	10.3	16.0	19.4	21.3	19.9	16.2	10.4	3.7	0.1
	2%	DB	-3.0	-0.9	5.2	15.3	24.6	28.3	28.6	25.8	21.0	12.9	3.3	-0.7
		MCWB	-3.7	-1.9	1.3	8.7	14.3	18.4	19.9	18.6	14.7	8.3	2.2	-1.3
	5%	DB	-5.2	-2.8	3.3	12.3	21.5	26.5	27.0	23.7	18.4	10.5	1.6	-2.6
		MCWB	-5.9	-3.7	0.5	6.7	13.1	17.9	19.2	17.5	13.4	7.6	0.8	-3.2
	10%	DB	-7.3	-5.0	1.6	9.5	18.4	24.4	25.3	21.8	15.8	8.1	0.3	-4.6
		MCWB	-7.9	-5.9	-0.5	4.9	11.5	17.1	18.4	16.7	12.0	5.8	-0.3	-5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	0.1	3.5	11.2	17.1	21.7	22.8	21.3	17.3	11.0	4.1	0.4
		MCDB	-0.9	0.9	7.2	17.3	25.7	26.9	28.3	25.9	22.4	13.5	4.7	0.6
	2%	WB	-3.6	-1.9	1.9	9.1	15.3	20.0	21.3	19.7	15.2	9.2	2.4	-1.3
		MCDB	-3.0	-1.0	4.3	14.0	22.3	25.9	26.6	24.2	19.8	12.2	3.2	-0.7
	5%	WB	-5.8	-3.7	0.8	7.2	13.8	18.7	20.2	18.4	13.9	7.6	0.9	-3.1
		MCDB	-5.2	-2.8	2.9	11.7	20.4	24.2	25.1	22.1	17.4	10.2	1.5	-2.6
	10%	WB	-7.8	-6.0	-0.4	5.3	12.2	17.7	19.2	17.2	12.4	5.9	-0.2	-5.2
		MCDB	-7.3	-5.0	1.6	9.1	17.6	23.2	23.9	20.8	15.4	7.8	0.4	-4.6
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.8	10.0	9.9	10.2	10.3	10.0	9.3	8.4	6.1	6.3	7.3
		MCDBR	7.0	6.6	10.0	13.5	14.6	14.2	12.7	12.8	13.2	9.8	5.3	7.4
		MCWBR	6.8	6.0	7.2	8.0	6.7	6.1	5.6	6.5	7.4	6.1	4.9	7.3
	5% WB	MCDBR	7.2	6.3	8.6	12.9	13.2	12.0	11.4	10.7	11.8	9.1	5.3	7.1
		MCWBR	7.1	6.0	6.6	8.0	6.8	5.9	5.9	6.1	7.0	6.2	5.2	7.1
Clear-Sky Solar Irradiance	taub		0.278	0.312	0.359	0.384	0.396	0.403	0.442	0.409	0.364	0.337	0.296	0.278
	taud		1.991	1.994	1.946	2.082	2.266	2.330	2.197	2.278	2.358	2.261	2.042	1.936
	Ebn at Noon		509	678	752	812	835	834	785	786	768	663	478	365
	Edh at Noon		57	93	131	135	121	116	130	111	88	72	54	44
All-Sky Solar Radiation	RadAvg		0.62	1.59	3.11	4.49	4.92	5.74	5.21	3.97	2.52	1.24	0.56	0.35
	RadStd		0.06	0.12	0.20	0.34	0.51	0.60	0.44	0.37	0.35	0.13	0.11	0.05

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page